04.

Simplification

Type - 5

1. Simplify: 0.8 + 0.08 + 0.008 + 8

(a) 880.8 (b) 8.888
(c) 8.808 (d) 8.088

RRB JE - 26/05/2019 (Shift-III)

Ans : (b) 0.8 + 0.08 + 0.008 + 8
= 8.888

2. 3.4 + 3.5 + 4.9 + 66 + 1.9 + 6.03 + 55 + 4.004 + 23 + 60 = ?

(a) 327.734 (b) 27.734
(c) 127.734 (d) 227.734

RRB RPF Constable -18/01/2019 (Shift-I)

Ans. (d) : Given,
3.4+3.5+4.9+66+1.9+6.03+55+4.004+23+60
= (66+55+23+60)+(3.4+3.5+4.9+1.9+6.03+4.004)
= 204 + 23.734
= 227.734

3. The value of 0.592 ÷ 0.8 = ?

(a) 7.4 (b) 0.74
(c) 740 (d) 0.074

RRB RPF-SI -13/01/2019 (Shift-II)

Ans : (b) From given expression,
0.592 ÷ 0.8

$$= \frac{0.592}{.8}$$

$$= \frac{592 \times 10}{8 \times 1000}$$

$$= \frac{74}{100}$$

= 0.74

4. 33.33 - 0.03 + 333.333 - 3.33 = ?

(a) 366.633 (b) 363.303
(c) 366.663 (d) 369.963

RRB RPF Constable -25/01/2019 (Shift-I)

Ans : (b) 33.33 - 0.03 + 333.333 - 3.33
$\Rightarrow$ 33.30 + 330.003 = 363.303

5. 6 kg 5 g = ?

(a) 6.05 kg (b) 0.65 kg
(c) 6.5 kg (d) 6.005 kg

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (d) 6kg 5g = $\left(6+\frac{5}{1000}\right)$kg

$= (6+0.005)$kg

= 6.005 kg

6. 5.52 – $(2.3)^2$ + $(0.8)^3$ × 0.12 ÷ $(0.4)^4$ – 3.14 = ?

(a) –0.51 (b) 0.42
(c) 0.51 (d) –0.63

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (a) 5.52 – $(2.3)^2$ + $(0.8)^3$ × 0.12 ÷ $(0.4)^4$ – 3.14
= 5.52 – 5.29 + 0.512 × 0.12 ÷ 0.0256 – 3.14
= 0.23 + 2.4 – 3.14
= – 0.51

7. 30.9 + 3.09 + 0.309 + 0.039 + 309 = ?

(a) 340.138 (b) 243.338
(c) 534.388 (d) 343.338

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (d) Given,
30.9 + 3.09 + 0.309 + 0.039 + 309
= 343.338

8. 7.892 + 35 – 0.005 + 10.345 = ?

(a) 53.232 (b) 52.232
(c) 23.322 (d) 0

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (a) Given,
= 7.892 + 35 – 0.005 + 10.345
= 7.892 + 10.345 + 35 – 0.005
= 53.237 – 0.005
= 53.232

9. 392 - 39.2 - 3.92 - 0.392 = ?

(a) 346.468 (b) 346.508
(c) 348.488 (d) 348.468

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (c) Given,
392 - 39.2 - 3.92 - 0.392
= 352.8 - 3.92 - 0.392
= 348.88 - 0.392
= 348.488

10. 0.098 + 0.98 + 9.8 + 98 = ?

(a) 108.338 (b) 108.428
(c) 108.878 (d) 108.378

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (c) Given,
0.098 + 0.98 + 9.8 + 98

$$= \frac{98}{1000} + \frac{98}{100} + \frac{98}{10} + 98$$

$$= \frac{98 + 980 + 9800 + 98000}{1000}$$

$$= \frac{108878}{1000}$$

= 108.878

11. $4 + 3 \times 4 + 3 \times 4^2 + 3 \times 4^3 + 3 \times 4^4 + 3 \times 4^5 = ?$

(a) 10×4^4 (b) 4^6
(c) 5×4^5 (d) 9×4^4

RRB Paramedical Exam – 20/07/2018 (Shift-II)

Ans : (b) Given,
$4+3\ (4 + 4^2 + 4^3 + 4^4 + 4^5)$
$\Rightarrow 4 + 3 \times 4\ (1 + 4 + 4^2 + 4^3 + 4^4)$
$\Rightarrow 4 + 3 \times 4 \left[\frac{4^5 - 1}{4 - 1}\right]$ $\{\because S_n = \frac{a(r^n - 1)}{r - 1}\}$
$\Rightarrow 4 + 3 \times 4 \times \frac{4^5 - 1}{3}$ (When r > 1)
$\Rightarrow 4 + 4^6 - 4$
$= 4^6$

12. Multiply 8795 by 125:

(a) 1099375 (b) 1099345
(c) 1099305 (d) 1098375

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (a) : 8795 × 125 = 8795 × (100 + 25)
= 879500 + 219875
= 10,99,375

13. Fill in the blanks with the most appropriate option.
395 – 39.5– 3.95 – 0.395 = ...

(a) 351.055 (b) 351.145
(c) 351.155 (d) 351.045

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (c) : 395 – 39.5– 3.95 – 0.395
= 395 – 43.45 – 0.395
= 395 – 43.845
= 351.155

14. 35 – 7.892 + 0.005 – 10.345 = ?

(a) 16.768 (b) 26.768
(c) 29.768 (d) 19.768

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (a) 35 - 7.892 + 0.005 – 10.345 = ?
= 35.005 – 18.237
= 16.768

15. The value of 20.9 + 2.09 + 0.209 + 0.029 + 29

(a) 62.228 (b) 52.228
(c) 42.228 (d) 52.00

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (b) : 20.9 + 2.09 + 0.209 + 0.029 + 29
= 52.228

16. 0.295 + 2.95 + 29.5 + 295 = ?

(a) 327.856 (b) 327.756
(c) 327.746 (d) 327.745

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (d) : 0.295 + 2.95 + 29.5 + 295 = ?
$\Rightarrow$ 3.245 + 324.5 = ?
? = 327.745

17. Calculate: 4237.43 + 453.32 + 24.12 – 387.23

(a) 4327.64 (b) 4646.64
(c) 4676.64 (d) 4587.64

RRB NTPC 28.04.2016 Shift : 2

Ans : (a) 4237.43 + 453.32 + 24.12 – 387.23
= 4714.87 – 387.23
= 4327.64

18. Simplify: $\frac{9}{13} \div \frac{18}{26} \div \frac{90}{52}$

(a) 45/26 (b) 13/45
(c) 26/45 (d) 45/13

RRB NTPC 03.04.2016 Shift : 1

Ans : (c) From given expression,
$\frac{9}{13} \div \frac{18}{26} \div \frac{90}{52}$
$= \frac{9}{13} \times \frac{26}{18} \times \frac{52}{90}$
$= \frac{26}{45}$

19. 5.16 × 3.2 = ?

(a) 15.502 (b) 16.512
(c) 17.772 (d) 17.52

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) Given expression,
5.16 × 3.2 = 16.512

20. 1. 1 + 12. 12 + 123. 123 = ?

(a) 134. 343 (b) 133.433
(c) 132.123 (d) 136.343

RRB NTPC 30.03.2016 Shift : 1

Ans : (d) From given expression,
1.1 + 12.12 + 123.123 = 136.343

21. 128-43+57-143+94 = ?

(a) 142 (b) 285
(c) 236 (d) 93

RRB NTPC 30.03.2016 Shift : 2

Ans : (d) From given expression,
128 – 43 + 57 -143 + 94
= 85 – 86 + 94 = 93

22. Find the value of 5231405 × 99999.

(a) 523135278595
(b) 723135268595
(c) 523135268595
(d) 623135268595

RRB NTPC 19.04.2016 Shift : 3

Ans : (c) 5231405 × 99999 = 523135268595

23. Solve: 7342015 × 9999?

(a) 73416544985 (b) 73455322985
(c) 73412807985 (d) 73412907985

RRB NTPC 12.04.2016 Shift : 2

Ans : (c) 7342015 × 9999
= 7342015 (10000-1)
= 73420150000 – 7342015
= 73412807985

24. Compute of 69696 × 9999 = ?

(a) 696980304 (b) 666890304
(c) 696809304 (d) 696890304

RRB NTPC 11.04.2016 Shift : 1

Ans : (d) From given expression,
69696 × 9999
= 69696 × (10000 –1)
= 696960000 – 69696
= 696890304

25. Calculate : 19170 ÷ 54 ÷ 5
(a) 17 (b) 1775
(c) 71 (d) 1757
RRB NTPC 11.04.2016 Shift : 3

Ans : (c) From given expression,
19170 ÷ 54 ÷ 5
= 355 ÷ 5
= 71

26. Calculate : 33333×9999
(a) 332396667 (b) 333297667
(c) 33329667 (d) 333296667
RRB NTPC 11.04.2016 Shift : 3

Ans : (d) 33333 × 9999 = 33333 (10000–1)
= 333330000 – 33333
= 333296667

27. Calculate :
0.00048 ÷ 0.08
(a) 0.06 (b) 0.006
(c) 0.0006 (d) 0.6
RRB NTPC 10.04.2016 Shift : 3

Ans : (b) From given expression,
0.00048 ÷ 0.08
$= \frac{0.00048}{0.08} = \frac{48\times100}{8\times100000} = \frac{6}{1000} = 0.006$

28. Calculate:
62160÷185÷24
(a) 41 (b) 8064
(c) 14 (d) 8046
RRB NTPC 07.04.2016 Shift : 1

Ans : (c) $= \frac{\frac{62160}{185}}{24} = \frac{62160}{185\times24} = 14$

29. Calculate:
66666 × 9999
(a) 665693334 (b) 666594334
(c) 666953334 (d) 666593334
RRB NTPC 07.04.2016 Shift : 1

Ans : (d) 66666 × 9999
= 66666 (10000 –1)
= 666660000 – 66666 = 666593334

30. 9876 + 34.567 – ? = 9908.221
(a) 23.45 (b) 234.6
(c) 2.345 (d) 2.346
RRB NTPC 06.04.2016 Shift : 2

Ans : (d) 9876 + 34.567 -? = 9908.221
9910.567 -? = 9908.221
? = 9910.567 – 9908.221
? = 2.346

31. Find the value of 1093 × 1093
(a) 1194649 (b) 1162481
(c) 1424649 (d) 1428481
RRB NTPC 22.04.2016 Shift : 1

Ans : (a) Given expression
1093 × 1093
= (1000 + 93) × 1093
= 1093000 + 1093 × 93
= 1093000 + 1093 × (100 – 7)
= 1093000 + 109300 – 1093 × 7
= 1093000 + 109300 – 1093 (10 – 3)
= 1093000 + 109300 – 10930 + 3279
= 1205579 – 10930 = 1194649

32. Calculate: 35968 ÷ 562 ÷ 8
(a) 80 (b) 512
(c) 8 (d) 521
RRB NTPC 22.04.2016 Shift : 2

Ans : (c) $\frac{\frac{35968}{562}}{8} = \frac{35968}{562}\times\frac{1}{8}$
$= \frac{4496}{562} = 8$

33. Calculate: 54367×9999
(a) 546315633
(b) 543655633
(c) 543651633
(d) 543615633
RRB NTPC 22.04.2016 Shift : 2

Ans : (d) 54367 × 9999
= 54367 × (10000 –1)
= 543670000–54367
= 543615633

34. $(64 \times 5^4) - (5^4 \times 16) = ?$
(a) 40,000 (b) 35,000
(c) 30,000 (d) 25,000
RRB NTPC 26.04.2016 Shift : 1

Ans : (c) $(64 \times 5^4) - (5^4 \times 16)$
= (64 × 625) – (625 × 16)
⇒ 40,000 – 10,000 = 30,000

35. The value of –261+(–380)–(–521)+821–(–121) is:
(a) 800 (b) 825
(c) 822 (d) 833
RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (c)
⇒ – 261 + (–380) – (–521) + 821 – (–121)
= – 261 – 380 + 521 + 821 + 121
= – 641 + 1463 = 822

Type - 5

36. Solve : $10^9 \times 10^2 \div 10^3$

(a) 10^8 (b) 10^6
(c) 10^2 (d) 10^5

RRB RPF Constable -22/01/2019 (Shift-III)

Ans : (a) $10^9 \times 10^2 \div 10^3 = \frac{10^9 \times 10^2}{10^3} = 10^6 \times 10^2$
$= 10^8$

37. $\{20-(25-33)\} \div \{-5\times4-(-6)\} + 56 \div (-27+13) = ?$

(a) -2 (b) -6
(c) -4 (d) 4

RRB RPF Constable -20/01/2019 (Shift-I)

Ans. (b)
$\{20-(25-33)\} \div \{-5\times4-(-6)\} + 56 \div (-27+13) = ?$
$= \{20+8\} \div \{-20+6\} + 56 \div (-14)$
$= \{28\} \div \{-14\} - 4$
$= -2 - 4 = -6$

38. {40 – (90 ÷ 5 × $\overline{16-8}$ ÷ 2 ÷ 3)} = ?

(a) 16 (b) 28
(c) 14 (d) 64

RRB RPF-SI -11/01/2019 (Shift-II)

Ans : (a) {40 – (90 ÷ 5 × $\overline{16-8}$ ÷ 2 ÷ 3)} = ?
⇒ {40 – (90 ÷ 5 × 8 ÷ 2 ÷ 3) } = ?
⇒ {40 – (90 ÷ 5 × 4 ÷ 3) } = ?
⇒ { 40 – (18 × 4 ÷ 3) } = ?
⇒ { 40 – 24} = ?
? = 16

39. 2 – [3 – { 6 – (5 – 4 – 3 + 10)}] = ?

(a) –3 (b) 1
(c) 2 (d) 4

RRB RPF Constable -19/01/2019 (Shift-II)

Ans : (a) From given expression,
2 – [3 – { 6 – (5 – 4 – 3 + 10)}] = ?
= 2 – [3–{6–(8)}]
= 2–[3–(–2)]
= 2–5
= –3

40. Solve the following.
2550 – [510 – {270 – (90 – 80 + 70)}]

(a) 2240 (b) 2230
(c) 2220 (d) 2210

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (b) = 2550 – [510 – {270 – (90 – 80 + 70)}]
= 2550 – [510 – {270 – 80}]
= 2550 – [510 – 190]
= 2550 – 320
= 2230

41. Solve the following :
23 – [23 – {23 – (23 – 23 + 23)}]

(a) –1 (b) 23
(c) 1 (d) 0

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (d) = 23 – [23 – {23 – (23 – 23 + 23)}]
= 23–[23–{23–23+23–23}]
= 23–[23–0]
= 23–23 = 0

42. Solve the following:
27 – [38 – {46 – (15 – 13 – 2)}]

(a) 35 (b) 31
(c) 29 (d) 30

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (a)
⇒ 27 – [38 – {46 – (15 – 13 – 2)}]
= 27 – [38– {46}]
= 27 – [38–46] = 27 – [–8]= 27 + 8 = 35

43. $56 \div \frac{1}{3}\left\{15+12-\left(9+6-\overline{5+7}\right)\right\} = ?$

(a) 9 (b) 8
(c) 12 (d) 7

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (d)
$56 \div \frac{1}{3}\left\{15+12-\left(9+6-\overline{5+7}\right)\right\}$
$= 56 \div \frac{1}{3}\left\{15+12-(15-12)\right\}$
$= 56 \div \frac{1}{3}\left\{15+12-3\right\}$
$= 56 \div \frac{1}{3}\left\{24\right\}$
$= 56 \div \frac{1}{3} \times 24$
$= 56 \div 8$
$= \frac{56}{8} = 7$

44. 77 ÷ [46 - {66 - (52 - 63 ÷ 9 × 3)}] = ?

(a) 5 (b) 6
(c) 7 (d) 11

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (c) $77 \div [46-\{66-(52-63\div9\times3)\}] = A$
$\Rightarrow 77 \div [46-\{66-(52-7\times3)\}] = A$
$\Rightarrow 77 \div [46-\{66-(31)\}] = A$
$\Rightarrow 77 \div [46-\{35\}] = A$
$\Rightarrow 77 \div [11] = A$
$\Rightarrow A = 7$

45. 140 ÷ [61 - {36 - (40 - 60 ÷ 12 × 6)}] = ?

(a) 2 (b) 5
(c) 4 (d) 10

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (c) From given expression,
$= 140 \div [61 - \{36 - (40 - 60 \div 12 \times 6)\}]$
$= 140 \div [61 - \{36 - (40 - 5 \times 6)\}]$
$= 140 \div [61 - \{36 - (10)\}]$
$= 140 \div [61 - 26]$
$= 140 \div 35 = 4$

46. (–5) {(20–(–2) × (–8)} = ?
(a) 180 (b) –20
(c) 20 (d) –180

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (b) From given expression,
(–5) {(20–(–2) × (–8)} = ?
= (–5) { 20–16)
= (–5) × 4
= –20

47. 78 – [5 + 3 of (25 – 2 × 10)] = ?
(a) 56 (b) 48
(c) 58 (d) 38

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (c) From given expression,
78 –[5 + 3 of (25 – 2 × 10)]
= 78 – [5 + 3 × (25 – 20)]
= 78 – [5 + 3 × 5]
= 78 – 20
= 58

48. 70 ÷ 5 × (10 – 8 ÷ 2) ÷ 3 = ?
(a) 7 (b) $\frac{1}{3}$
(c) 3 (d) 28

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (d) From given expression,
70 ÷ 5 × (10 – 8 ÷ 2) ÷ 3 = ?
70 ÷ 5 × (10 – 4) ÷ 3 = ?
70 ÷ 5 × 6 ÷ 3 = ?
70 ÷ 5 × 2 = ?
14 × 2 = ?
? = 28

49. $\left[\left\{2\frac{1}{3} - (5 + (2 - 3)\right\} + 3\frac{1}{2}\right] = ?$
(a) $\frac{11}{2}$ (b) $\frac{12}{6}$
(c) $\frac{11}{6}$ (d) 2

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (c) From given expression,
$\left[\left\{2\frac{1}{3} - (5 + (2 - 3)\right\} + 3\frac{1}{2}\right] = ?$
$\left[\left\{2\frac{1}{3} - (5 + (2 - 3)\right\} + 3\frac{1}{2}\right]$
$= \left[\left\{\frac{7}{3} - (5 - 1)\right\} + \frac{7}{2}\right]$
$= \left[\left\{\frac{7}{3} - 4\right\} + \frac{7}{2}\right]$
$= \left[\frac{-5}{3} + \frac{7}{2}\right]$
$= \left[\frac{-10 + 21}{6}\right]$
$= \frac{11}{6}$

50. 63 – (- 3) (-2 -8 -4) ÷ 3 of { 5 + (-2) (-1) } = ?
(a) -60 (b) 60
(c) 65 (d) 61

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (d) 63 – (–3) (–2 –8 – 4) ÷ 3 of { 5 + (–2) (–1)}
= 63 – (–3) (–14) ÷ 3 × { 5 + 2}
= 63 – (–3) (–14) ÷ 3 × 7
= 63 – (–3) (–14) ÷ 21
= 63 – 42 ÷ 21
= 63 – 2
= 61

51. 72 ÷ [27 – {35 – (42 – 45 ÷ 9 × 2) }] = ?
(a) 3 (b) 8
(c) 6 (d) 4

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (a) From given expression,
72 ÷ [27 – { 35 – (42 – 45 ÷ 9 × 2)}]
= 72 ÷ [27 – { 35 – (42 – 10)}]
= 72 ÷ [27- { 35 – 32 }]
= 72 ÷ [27 – 3]
= 72 ÷ 24
= 3

52. If T = (93 + 15) ÷(3 × 4) – 24 + 8, then what will be the value of T?
(a) –4 (b) –7
(c) –2 (d) –5

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (b) Given,
T = (93 + 15) ÷ (3 × 4) – 24 + 8
= (108) ÷ (12) – 24 + 8
= 108 ÷ 12 – 24 + 8
= 9 – 24 + 8
= 17 – 24
T = – 7

53. 75 ÷ [35 – {63 – (79 – 54 ÷ 9 × 6)}] = ?
(a) 5 (b) 3
(c) 15 (d) 25

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (a) From given expression,
75 ÷ [35 – {63 – (79 – 54 ÷ 9 × 6)}] = ?
⇒ 75 ÷ [35 – {63 – (79 – 6 × 6)}] = ?
⇒ 75 ÷ [35 – {63 – (79 – 36}] = ?
⇒ 75 ÷ [35 – {63 – 43}] = ?
⇒ 75 ÷ [35 – 20] = ?
⇒ 75 ÷ 15 = 5

54. 0.36 + 0.284 ÷ 0.4 × 0.8 – 0.038 = ?

(a) 0.548 (b) 0.89
(c) 1.25 (d) 1.2

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (b) From given expression,

$0.36 + 0.284 \div 0.4 \times 0.8 - 0.038$
$\Rightarrow 0.36 + 0.71 \times 0.8 - 0.038$
$\Rightarrow 0.36 + 0.568 - 0.038$
$\Rightarrow 0.928 - 0.038 = 0.89$

55. $3+\left[32\div 8\times 52\div(4+9)\right]=?$

(a) 19 (b) 20
(c) 18 (d) 21

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : From given expression,

$3+\left[32\div 8\times 52\div(4+9)\right]=?$
$=3+\left[4\times 52\div 13\right]$
$=3+\frac{4\times 52}{13}$
$=3+4\times 4=3+16=19$

56. $144\div\left[40-\left\{37-\left(25-112\div\overline{7\times 4}\right)\right\}\right]$

(a) 4 (b) 8
(c) 2 (d) 6

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (d) : From given expression,

$144\div\left[40-\{37-(25-112\div\overline{7\times 4})\}\right]$
$\Rightarrow 144\div\left[40-\{37-(25-112\div 28)\}\right]$
$\Rightarrow 144\div\left[40-\{37-(25-4)\}\right]$
$\Rightarrow 144\div\left[40-\{37-21\}\right]$
$\Rightarrow 144\div\left[40-16\right]$
$\Rightarrow 144\div 24=6$

57. If G = (96 ÷12) + 14 × (12 +8) ÷ 2, then what will be 1/4th of G?

(a) 148 (b) 37
(c) 36 (d) 38

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (b) Given,

G = (96 ÷ 12) + 14 × (12 + 8) ÷ 2
= 8 + 14 × 20 ÷ 2
= 8 + 14 × 10
= 8 + 140
$\boxed{G=148}$

Then 1/4th of G,

$\frac{G}{4}=\frac{148}{4}=37$

58. $18\div\frac{1}{8}\{11+16-(10+7-\overline{6+8})\}=?$

(a) 6 (b) 9
(c) 18 (d) 3

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (a) From given expression,

$?=18\div\frac{1}{8}\{11+16-(10+7-\overline{6+8})\}$
$=18\div\frac{1}{8}\{11+16-(10+7-14)\}$
$=18\div\frac{1}{8}\{11+16-3\}$
$=18\div\frac{1}{8}\times 24$
$?=18\div 3=6$

59. $22-\left(\frac{1}{4}\right)\{-5-(-48)\div(-16)\}=?$

(a) 21 (b) 22
(c) 20 (d) 24

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (d) From given expression,

$22-\frac{1}{4}\{-5-(-48)\div(-16)\}=?$
$=22-\frac{1}{4}\{-5-3\}$
$=22+\frac{1}{4}\times 8=22+2=24$

60. What is the value of
$(-45+7\times 23-(247\div 13)-11)\div 2$?

(a) –47 (b) –14
(c) 43 (d) 86

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (c) From given expression,

(–45 + 7 × 23 – (247 ÷ 13) – 11) ÷ 2
= (–45 + 161 – 19 – 11) ÷ 2
= (161 – 75) ÷ 2
= 86 ÷ 2
= 43

61. $74-[85\div\{49-(41-3^5\div\overline{9\times 3})\}]=?$

(a) 59 (b) 79
(c) 49 (d) 69

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (d) From given expression,

$74-[85\div\{49-(41-3^5\div\overline{9\times 3})\}]$
$?=74-[85\div\{49-(41-3^5\div 27)\}]$
$=74-[85\div\{49-(41-243\div 27)\}]$
$=74-[85\div\{49-(41-9)\}]$
$=74-[85\div\{49-32\}]$
$=74-[85\div\{17\}]$
$=74-[85\div 17]$
$=74-[5]$
$?=74-5=69$

62. (12 - 1) of (16 + 15) × (119 + 113 - 16) = ?

(a) 73656 (b) 73660
(c) 73600 (d) 73650

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (a) From given expression,
(12 - 1) of (16 + 15) × (119 + 113 - 16)
= 11 of 31 × 216
= 11 × 31 × 216
= 73656

63. Sumplify:
25+15–(51)+(4×15 of 17) ÷ 20+ $\overline{6-2}$ = ?

(a) 45 (b) 44
(c) –44 (d) –45

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (b) From given expression,
25+15–(51)+(4×15 of 17) ÷ 20+ $\overline{6-2}$

$\Rightarrow \quad 25+15-51+\left(\frac{4\times15\times17}{20}\right)+4$

$\Rightarrow \quad 40-51+51+4 = 44$

64. Find the following:

$$10+\left\{26-15\times\left(20-5\div2\times\overline{7-5}\right)\right\}=?$$

(a) 189 (b) −198
(c) 198 (d) −189

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (d) : From given expression,

$$10+\left\{26-15\times\left(20-5\div2\times\overline{7-5}\right)\right\}=?$$

$$=10+\left\{26-15\times\left(20-5\times\frac{1}{2}\times2\right)\right\}$$

$$=10+\{26-15\times(20-5)\}$$

$$=10+\{26-15\times(15)\}$$

$$=10+\{26-225\}$$

$$=10+26-225$$

$$\boxed{=-189}$$

65. Solve: 6.9–[(3.19×0.7)–0.5 of (8.5-3.04)–2.85]

(a) –4.36 (b) 10.247
(c) 6.61 (d) 8.79

RRB NTPC 29.04.2016 Shift : 1

Ans :(b) From given expression,
6.9 – [(3.19×0.7)- 0.5 of (8.5-3.04)-2.85]

$$=6.9-[2.233-0.5\times5.46-2.85]$$

$$=6.9-[2.233-2.73-2.85]$$

$$=6.9-[-3.347]$$

$$=6.9+3.347=10.247$$

66. Solve: (50 + 0.5 × 20) ÷ 0.7

(a) 8.571 (b) 857.1
(c) 85.71 (d) 72.85

RRB NTPC 02.04.2016 Shift : 2

Ans : (c) From given expression,
(50 + 0.5 × 20) ÷ 0.7
= (50 + 10) ÷ 0.7
= 60 ÷ 0.7
= 85.71

67. Simplify: $\left[2\frac{1}{3}-1\frac{1}{2}\right]$ of $\frac{3}{5}+1\frac{2}{5}\div2\frac{1}{3}$

(a) $\frac{1}{10}$ (b) $\frac{3}{10}$
(c) $1\frac{1}{10}$ (d) 1

RRB NTPC 29.03.2016 Shift : 1

Ans : (c) From given expression,

$$\left[2\frac{1}{3}-1\frac{1}{2}\right]\text{of}\,\frac{3}{5}+1\frac{2}{5}\div2\frac{1}{3}$$

$$=\left[\frac{7}{3}-\frac{3}{2}\right]\times\frac{3}{5}+\frac{7}{5}\div\frac{7}{3}$$

$$=\frac{14-9}{6}\times\frac{3}{5}+\frac{7}{5}\times\frac{3}{7}$$

$$=\frac{5}{6}\times\frac{3}{5}+\frac{3}{5}=\frac{1}{2}+\frac{3}{5}=\frac{5+6}{10}=\frac{11}{10}=1\frac{1}{10}$$

68. Calcuplate : 4082 ÷ 157 – 23

(a) - 3 (b) 3
(c) 2041/67 (d) 2014/67

RRB NTPC 11.04.2016 Shift : 2

Ans : (b) From given expression,
4082 ÷ 157 – 23
4082 ÷ 157 – 23
26 – 23
= 3

69. $4+\frac{1}{6}\times[\{-12\times(24-13-3)\}\div(20-4)]=?$

(a) 4 (b) 6
(c) 5 (d) 3

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) From given expression,

$$4+\frac{1}{6}\times\left[\{-12\times(24-13-3)\}\div(20-4)\right]$$

$$=4+\frac{1}{6}\times\left[\{-12\times8\}\div16\right]$$

$$=4+\frac{1}{6}\times(-6)=4-1$$

$$=3$$

70. The value of 14 ÷ {(5 of 2 – 3)} × 4 (7 – 2) is:

(a) $\frac{1}{10}$ (b) 40
(c) 44 (d) $\frac{14}{19}$

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (b) From given expression,

$$14 \div \{(5\times2-3)\}\times4(7-2)$$
$$=14\div\{(10-3)\}\times4(7-2)$$
$$=14\div7\times4\times5$$
$$=2\times4\times5=40$$

71. 45 – [38 – {80 ÷ 4 – (8 – 12 ÷ 3) ÷ 4}] = ?

(a) 25 (b) 27
(c) 26 (d) 28

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (c) From given expression,

45 – [38 – {80 ÷ 4 – (8 – 12 ÷ 3) ÷ 4}]
= 45–[38–{80÷4–4÷4}]
= 45–[38–{20–1}]
= 45–[38–19]
= 45–19= 26

72. Solve the following:

$$72\div\frac{1}{2}\{15+12-(9+6-\overline{5+7})\}=?$$

(a) 6 (b) 9
(c) 12 (d) 8

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (a) From given expression,

$$72\div\frac{1}{2}\{15+12-(9+6-\overline{5+7})\}=?$$
$$=72\div\frac{1}{2}\{15+12-(9+6-12)\}$$
$$=72\div\frac{1}{2}\{15+12-(15-12)\}$$
$$=72\div\frac{1}{2}\{15+12-3\}=72\div\frac{1}{2}\{27-3\}$$
$$=72\div\frac{1}{2}\times24=72\div12=6$$

73. Find the value of the given expression :
6–36 × 3 ÷ 6 + 5 = ?

(a) $\frac{42}{11}$ (b) 7
(c) $\frac{-42}{11}$ (d) –7

RRB ALP & Tec. (30-08-18 Shift-II)

Ans : (d) From given expression,

6–36 × 3 ÷ 6 + 5

$$=6-36\times\frac{3}{6}+5,\quad =6-36\times\frac{1}{2}+5$$
$$=6-18+5,\quad =-7$$

74. Solve the following:
(–6) [40 ÷ {7 – (–3)}] = ?

(a) 24 (b) –60
(c) 60 (d) –24

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (d) From given expression,

(–6) [40 ÷ {7 – (–3)}] = ?
(–6) [40 ÷ {10}]
(–6) [40 ÷ 10]
(–6) [4] = –24

75. 4 + (1/6) [{-10 × (25 - 13 - 3)} ÷ (-5)] = ?

(a) 8 (b) 9
(c) 6 (d) 7

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (d) From given expression,

$$4+\frac{1}{6}\left[\{-10\times(25-13-3)\}\div(-5)\right]$$
$$\Rightarrow 4+\frac{1}{6}\left[\{(-10\times9)\div(-5)\}\right]$$
$$\Rightarrow 4+\frac{1}{6}\left[(-90)\times\left(\frac{1}{-5}\right)\right]$$
$$\Rightarrow 4+\frac{1}{6}[18]\Rightarrow 4+3=7$$

76. Solve the following :
36 – [18 – {14 – (15 – 4 ÷ 2 × 2)}]

(a) 20 (b) 22
(c) 21 (d) 23

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (c) $36-[18-\{14-(15-4\div2\times2)\}]$

$$=36-[18-\{14-(15-2\times2)\}]$$
$$=36-[18-\{14-11\}]$$
$$=36-[18-3]$$
$$=36-15$$
$$=21$$

77. Solve the following :
24 ÷ (19–9 ÷ 3 × 9) = ?

(a) –3 (b) –4
(c) 3 (d) 6

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (a) From given expression,

$$24\div(19-9\div3\times9)$$
$$=24\div(19-3\times9)$$
$$=24\div(19-27)$$
$$=24\div(-8)$$
$$=-3$$

78. What is the value of 12 + 3 (–2 × 3) – (18÷ 6)?

(a) 5 (b) –9
(c) –5 (d) 9

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (b) From given expression,

12 + 3 (–2 × 3) – (18 ÷ 6)
= 12 + 3 (–6) – 3
= 12 – 18 – 3
= 12 – 21 = – 9

79. $25-\frac{1}{2}\{5+4-(3+2-1+3)\}=?$

(a) 23 (b) 21
(c) 24 (d) 22

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (c) From given expression,

$=25-\frac{1}{2}\{5+4-(3+2-1+3)\}$

$=25-\frac{1}{2}\{5+4-7\}$

$=25-\frac{1}{2}\times 2=24$

80. $23-\left[24-\left\{25-\left(26-\overline{27-28}\right)\right\}\right]=?$

(a) –2 (b) –3
(c) –1 (d) 1

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) From given expression,

$23-\left[24-\{25-(26-\overline{27-28})\}\right]$

$=23-\left[24-\{25-(26+1)\}\right]$

$=23-\left[24-\{25-27\}\right]$

$=23-\left[24+2\right]$, $=23-26=-3$

81. **(–8) [36 ÷ {7–(–2)}] ÷ (–4) {19–(–3) × (–5)} = ?**

(a) 2 (b) –4
(c) 4 (d) –2

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (a) From given expression,

$=(-8)\left[36\div\{7-(-2)\}\right]\div(-4)\{19-(-3)\times(-5)\}$

$=(-8)\left[36\div\{9\}\right]\div(-4)\{19-15\}$

$=(-8)[4]\div(-16)$

$=\frac{32}{16}=2$

82. **Solve the following:**

45 – [38 – {60 ÷ 3 – (6 – 9 ÷ 3) ÷ 3}]

(a) 25 (b) 26
(c) 24 (d) 21

RRB ALP & Tec. (13-08-18 Shift-III)

Ans : (b) From given expression,

45 – [38 – {60 ÷ 3 – (6 – 9 ÷ 3) ÷ 3}]
= 45 – [38 – {20 – 3÷ 3}]
= 45 – [38 – 19]
= 45–19 = 26

83. **Solve the following.**

$\left\{38-(60\div 5\times\overline{16-8}\div 2\div 3)\right\}=?$

(a) 30 (b) 29
(c) 22 (d) 37

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (c) From given expression,

$\left\{38-\left(60\div 5\times\overline{16-8}\div 2\div 3\right)\right\}=?$

$=\left\{38-\left(60\div 5\times 8\div 2\div 3\right)\right\}$

$=\left\{38-\left(60\div 5\times 4\div 3\right)\right\}$

$=\left\{38-\left(12\times\frac{4}{3}\right)\right\}$

$=\{38-16\}=22$

84. **Solve the following:**

22 – (1/4) {–5 – (–48) ÷ (–16)}

(a) 0 (b) 24
(c) 22 (d) 21

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (b) From given expression,

$22-\frac{1}{4}\left\{-5+48\times\left(\frac{-1}{16}\right)\right\}$

$=22+2=24$

85. **What is the value of |3(1) – 6|?**

(a) 3 (b) 0
(c) –3 (d) 4

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (a) = |3(1)–6|
= |3–6|= 3
(Mod always gives the positive value.)

Type - 3

86. **If 2508 ÷ 12.54 + (X × 11) = 200, Then find the value of X.**

(a) 2.5 (b) 0
(c) 4 (d) 3.5

RRB JE - 22/05/2019 (Shift-III)

Ans : (b) Given,

2508 ÷ 12.54 + (X × 11) = 200
200 + (X × 11) = 200
(X × 11) = 200 – 200

$X=\frac{0}{11}$

X = 0

87. **If** $\frac{x}{\sqrt{128}}=\frac{\sqrt{162}}{x}$ **Then find the value of 'X'.**

(a) 14 (b) 12
(c) 144 (d) 13

RRB JE - 23/05/2019 (Shift-III)

Ans : (b) Given,

$\frac{x}{\sqrt{128}}=\frac{\sqrt{162}}{x}$

$\Rightarrow x^2=\sqrt{162}\times\sqrt{128}$

$\Rightarrow x^2=\sqrt{18\times 9\times 16\times 8}$

$\Rightarrow x^2=144$

$\Rightarrow x=\sqrt{144}$

x = 12

88. If $\frac{x}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$, then find the value of A and B.

(a) 2, –2 (b) 1/2, –1/2
(c) 1/2, 1/2 (d) 2, 2

RRB JE - 23/05/2019 (Shift-III)

Ans : (c)

$$\frac{x}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$\frac{x}{x^2-1} = \frac{A(x+1)+B(x-1)}{(x^2-1)}$$

x = Ax + A + Bx – B
x + 0 = x (A + B) + (A – B)
A + B = 1 (On comparing)
A – B = 0
2A = 1
A = 1/2
B = 1/2

89. If 5x/(1 + 1/(1 + x/(1 – x))) =1, find the value of 'x'.

(a) 1 (b) 1/3
(c) 5/3 (d) 2/3

RRB JE - 29/05/2019 (Shift-II)

Ans : (b) Given,

$$= \frac{5x}{1+\frac{1}{1+\frac{x}{1-x}}} = \frac{5x}{1+\frac{(1-x)}{1-x+x}} = \frac{5x}{1+\frac{(1-x)}{1}} = \frac{5x}{2-x} = 1$$

$\therefore 5x = 2 - x$
$6x = 2$
$x = \frac{1}{3}$

90. If 138.019 + 341.981 – 146.395 = 133.605 + a then find the value of 'a'.

(a) 248 (b) 120.085
(c) 200 (d) 295.5

RRB JE - 30/05/2019 (Shift-II)

Ans : (c) Given,
138.019 + 341.981 – 146.395 = 133.605 + a
= 138.019 + 341.981 = 146.395 + 133.605 + a
= 480 = 280 + a
a = 480 – 280
a = 200

91. If |–4 x + 4| – 6 = – 6, then find the value of 'x'.

(a) – 1 (b) 0
(c) 1 (d) 2

RRB JE - 30/05/2019 (Shift-II)

Ans : (c) From given expression,
|–4x + 4| – 6 = –6
|–(4x – 4)| – 6 = –6
4x – 4 = –6 + 6
4x – 4 = 0
x = 1

92. If 1.5x=0.04y then find the value of (y–x)/(y+x)

(a) 73/77 (b) 0.73/77
(c) 73/770 (d) 730/77

RRB JE - 28/05/2019 (Shift-II)

Ans : (a) From given expression,
1.5x = 0.04y

$$\frac{x}{y} = \frac{.04}{1.5}$$

$$\frac{x}{y} = \frac{4}{150}$$

$$\frac{x}{y} = \frac{2}{75}$$

Let x = 2k , y = 75k

$$\frac{y-x}{x+y} = \frac{75k-2k}{75k+2k} = \frac{73k}{77k} = \frac{73}{77}$$

93. If (0.29 + 0.25 + 0.01)/ 0.005 = a%; then find the value of 'a'.

(a) 5000 (b) 500
(c) 11000 (d) 110

RRB RPF Constable -17/01/2019 (Shift-III)

Ans : (c) given,

$$\frac{(0.29+0.25+0.01)}{0.005} = a\%$$

$$\frac{0.55}{0.005} = \frac{a}{100}$$

$$\frac{55\times10^{-2}}{5\times10^{-3}} = \frac{a}{100}$$

a = 11000

94. If 49×17 = 833, then what is the value of 0.0833 ÷ 4.9

(a) 1.7 (b) 0.017
(c) 0.0017 (d) 0.17

RRB RPF-SI -12/01/2019 (Shift-III)

Ans : (b) 49 × 17 = 833
0.0833 ÷ 4.9 = ?

$$? = \frac{0.0833}{4.9} = 0.017$$

95. If 185 × 28 = 5180, then what is the value of $\frac{51.8}{18.5}$?

(a) 0.28 (b) 280
(c) 2.8 (d) 28

RRB RPF-SI -16/01/2019 (Shift-III)

Ans : (c) Given,
185× 28 = 5180

$$28 = \frac{5180}{185}$$

Then,

$$28 = \frac{518}{185}\times10$$

$\frac{28}{10} = \frac{518}{185}$

$2.8 = \frac{518}{185}$

अत: $\frac{51.8}{18.5} = 2.8$

96. Simplify: $y+[y-(y+x)+\{y-(y-x)\}+(z+x)] = ?$

(a) y (b) $x+y+z$

(c) $y+z$ (d) $x+z$

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (b) From given expression,

$y + [y-(y+x) + \{y-(y-x)\} + (z+x)]$

$= y + [y - y - x + \{y - y + x\} + z + x]$

$= y - x + x + z + x$

$= x + y + z$

97. 23 × 31 = 713, then 0.00713 ÷ 3.1 = ?

(a) 0.023 (b) 0.0023

(c) 0.23 (d) 2.3

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (b) Given,

$23 \times 31 = 713$

$23 = \frac{713}{31}$ .....(I)

Then,

$0.00713 \div 3.1 = \frac{713 \times 10}{31 \times 100000}$

$= 23 \times \frac{1}{10000} = 0.0023$ .....$\left(From\ eq^n\ (I)\right)$

98. If $\frac{0.5-0.1x}{1.3-0.8x} = 0.2$, then x = ?

(a) – 1 (b) – 1

(c) – 3 (d) – 4

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (d) Given,

$\frac{0.5-0.1x}{1.3-0.8x} = 0.2,$

$0.5 - 0.1x = 0.26 - 0.16x$

$0.16x - 0.1x = 0.26 - 0.5$

$0.06x = -0.24$

$x = -\frac{0.24}{0.06}$

$x = -4$

99. If 31 × 23 = 713, then 310 × 0.023 = ?

(a) 0.713 (b) 71.3

(c) 7.13 (d) 0.0713

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (c) Given,

$31 \times 23 = 713$

Then, $310 \times 0.023 = ?$

$\Rightarrow 31 \times 10 \times \frac{23}{1000}$

$\Rightarrow \frac{31 \times 23}{100} = \frac{713}{100} = 7.13$

100. If 31 × 23 = 713, then what is the value of 3100 × 0.00023 ?

(a) 71.3 (b) 7.13

(c) 0.713 (d) 0.0713

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) Given,

$31 \times 23 = 713$

Then,

$3100 \times 0.00023 = 31 \times 100 \times 23 \times \frac{1}{100000}$

$= 31 \times 23 \times \frac{1}{1000} = 713 \times \frac{1}{1000} = 0.713$

101. Solve the following equation to find the value of x.

$\frac{(x-5)}{3} - \frac{(x-2)}{4} = \frac{7}{2}$

(a) 42 (b) 60

(c) 56 (d) 52

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c) Given,

$\frac{(x-5)}{3} - \frac{(x-2)}{4} = \frac{7}{2}$

$\Rightarrow \frac{4(x-5)-3(x-2)}{12} = \frac{7}{2}$

$\Rightarrow \frac{4x-20-3x+6}{12} = \frac{7}{2}$

$\Rightarrow \frac{x-14}{12} = \frac{7}{2}$

$\Rightarrow x - 14 = 42$

$\Rightarrow x = 56$

102. 23 × 33 = 759, then what is 0.00759 ÷ 3.3 ?

(a) 0.023 (b) 0.0023

(c) 2.3 (d) 0.23

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b) Given,

$23 \times 33 = 759$

$23 = \frac{759}{33}$

Then,

$0.00759 \div 3.3$

$= \frac{759 \times 10}{33 \times 100000} = 23 \times \frac{1}{10000} = 0.0023$

103. If 1120/√x = 80, then x = ?

(a) 225 (b) 196

(c) 125 (d) 336

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (b) Given,

$1120/\sqrt{x} = 80$

$\frac{1120}{\sqrt{x}} = 80$

$\frac{1120}{80} = \sqrt{x}$

$14 = \sqrt{x}$

$x = 14^2 = 196$

104. If p = 36 – 2 (20 + 12 ÷ 4 × 3 – 2 × 2) + 10, then what is twice of p?

(a) –8 (b) –4
(c) –2 (d) –10

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (a) Given,

P = 36– 2 (20 + 12 ÷ 4 × 3 – 2 × 2) + 10
P = 36 – 2 (20 + 3 × 3 – 2 × 2) + 10
P = 36–2 (29 – 4) + 10
P = 36 – 50 + 10
P = –4

Hence, twice of p will be = – 4 × 2 = –8

105. If 1131 ÷ 39 = 29, then 11.31 ÷ 0.0029 = ?

(a) 3.9 (b) 3900
(c) 390 (d) 0.39

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (b) Given,

$1131 \div 39 = 29$

$\frac{1131}{39} = 29$

$\frac{1131}{29} = 39$

Then,

$11.31 \div 0.0029 = \frac{11.31}{0.0029}$

$= \frac{1131 \times 10000}{29 \times 100} = 39 \times 100$

$= 3900$

106. If $\frac{1}{x-a-b} = \frac{1}{x} - \frac{1}{a} - \frac{1}{b}$ then find the value of x.

(a) –a, b (b) a, b
(c) a, –b (d) –a, –b

RRB Group-D – 15/11/2018 (Shift-I)

Ans : (b) Given,

$\frac{1}{x-a-b} = \frac{1}{x} - \frac{1}{a} - \frac{1}{b}$

$\frac{1}{a} + \frac{1}{b} = \frac{1}{x} - \frac{1}{x-a-b}$

$\frac{a+b}{ab} = \frac{x-a-b-x}{x(x-a-b)}$

$\frac{a+b}{ab} = \frac{-(a+b)}{x(x-a-b)}$

$x^2 - (a + b)x + ab = 0$
$x(x - a) - b(x - a) = 0$
$(x - a)(x - b) = 0$
$x = a, x = b$

So, the value of x is a and b.

107. If $\sqrt{75} + \sqrt{363} = \sqrt{N}$, then what is the valeu of N?

(a) 729 (b) 438
(c) 768 (d) 27

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (c) From given expression,

$\sqrt{75} + \sqrt{363} = \sqrt{N}$

On squaring both sides,

$\left(\sqrt{75}\right)^2 + \left(\sqrt{363}\right)^2 + 2\sqrt{75} \times \sqrt{363} = \left(\sqrt{N}\right)^2$

$75 + 363 + 2 \times 5\sqrt{3} \times 11\sqrt{3} = N$

$438 + 330 = N$

$N = 768$

108. If 202.4 ÷ x = 5.06, then find the value of x.

(a) 30 (b) 42
(c) 43 (d) 40

RRB Group-D – 31/10/2018 (Shift-III)

Ans : (d) From given expression,

$202.4 \div x = 5.06$

$\Rightarrow \frac{202.4}{x} = 5.06$

$\Rightarrow x = \frac{202.4}{5.06}$

$\Rightarrow x = \frac{20240}{506}$

$x = 40$

109. Evaluate:

$\sqrt{19600} + \sqrt{0.0196} + \sqrt{0.00000196}$

(a) 142.1414 (b) 140.1414
(c) 143.1414 (d) 141.1414

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (b) : From given expression,

$\sqrt{19600} + \sqrt{0.0196} + \sqrt{0.00000196}$

$= 140 + 0.14 + 0.0014$

$= 140.1414$

110. What is the value of x?

$\frac{144}{1.44} = \frac{14.4}{x}$

(a) 0.144 (b) 1.44
(c) 0.00144 (d) 0.0144

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (a) From given expression,

$\frac{144}{1.44} = \frac{14.4}{x}$

$\frac{144 \times 100}{144} = \frac{144}{x \times 10}$ or $x = \frac{144 \times 144}{144 \times 100 \times 10}$

$x = 0.144$

111. If $123 \times 356 = 43788$, then $1.23 \times 35.6 = ?$

a) 4.3788 (b) 43.788
(c) 0.43788 (d) 437.88

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (b) Given,

$123 \times 356 = 43788$

$1.23 \times 35.6 = \frac{123}{100} \times \frac{356}{10} = \frac{43788}{1000} = 43.788$

112. Find the value of 'a'?

$\frac{7}{a-2} = \frac{5}{a+4}$

(a) –19 (b) 38
(c) 19 (d) –38

RRB NTPC 19.01.2017 Shift : 2

Ans : (a) From given expression,

$\frac{7}{a-2} = \frac{5}{a+4}$

$7a + 28 = 5a - 10$

$2a = -38$

$a = -19$

113. Simplify:

$1 \div [\{p^2 / (p + 6)\} + \{6p/(p + 6)\}]$

(a) $\frac{1}{p}$ (b) $\frac{1}{(p+6)}$
(c) $p + 6$ (d) p

RRB NTPC 17.01.2017 Shift-2

Ans : (a) From given expression,

$1 \div \left[\{p^2/(p+6)\} + \{6p/(p+6)\}\right]$

$= 1 \div \left[\frac{p^2}{p+6} + \frac{6p}{p+6}\right]$

$= 1 \div \left[\frac{p(p+6)}{(p+6)}\right]$

$= 1 \div p = \frac{1}{p}$

114. If $\frac{x}{5} + \frac{y}{7} = 1$ and x = 2, then y = ?

(a) 3/5 (b) 7/5
(c) 5/3 (d) 21/5

RRB N TPC 31.03.2016 Shift : 1

Ans : (d) Given,

$\frac{x}{5} + \frac{y}{7} = 1$

$\because x = 2$

$\therefore \frac{2}{5} + \frac{y}{7} = 1$

$\frac{y}{7} = 1 - \frac{2}{5}$

$\frac{y}{7} = \frac{3}{5}$

$y = \frac{21}{5}$

115. If $4^3 \times 5^6 = 10^m$, then find the value of m.

(a) 4 (b) 6
(c) 8 (d) 2

RRB NTPC 29.03.2016 Shift : 2

Ans : (b) From given expression,

$4^3 \times 5^6 = 10^m$

$(4 \times 5)^3 \times 5^3 = 10^m$

$20^3 \times 125 = 10^m$

$8000 \times 125 = 10^m \Rightarrow 10^m = 1000000$

$\Rightarrow 10^m = 10^6$

$\Rightarrow \boxed{m = 6}$

116. If 4515 ÷17.5 = 258, then 45.15 ÷ 1.75 = ?

(a) 2.58 (b) 0.258
(c) 25.8 (d) 258

RRB NTPC 09.04.2016 Shift : 3

Ans : (c) Given,

$4515 \div 17.5 = 258$

$\frac{4515}{17.5} = 258$

$\frac{45.15 \times 100}{1.75 \times 10} = 258$

$\frac{45.15}{1.75} = 258 \times \frac{10}{100}$

$45.15 \div 1.75 = 25.8$

117. 19 × 23 = 437. Then 190 × 0.023 = ?

(a) 0.0437 (b) 0.437
(c) 43.7 (d) 4.37

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (d) Given,

$19 \times 23 = 437$

So,

$190 \times 0.023 = 19 \times 23 \times \frac{10}{1000}$

$= 437 \times \frac{10}{1000} = 4.37$

Type - 4

118. Simplify : (−4.6)× (−4.6) ÷ (−4.6 + 0.6)

(a) - 5.29 (b) - 0.529
(c) - 4.06 (d) 5.01

RRB RPF Constable -24/01/2019 (Shift-I)

Ans : (a) From given expression,

(-4.6) × (-4.6) ÷ (-4.6 + 0.6)

= (-4.6) × (-4.6) ÷ (- 4.0)

= (-4.6) × (-4.6) ×1/(-4) = -5.29

119. Which of the following will take place of *?

$1\frac{1}{4} + 1\frac{1}{6} - 1\frac{1}{8} = * + 1\frac{1}{12}$

(a) 5/12 (b) 7/12
(c) 7/24 (d) 5/24

RRB JE - 22/05/2019 (Shift-III)

Ans : (d) From given expression,

$$1\frac{1}{4}+1\frac{1}{6}-1\frac{1}{8}=*+1\frac{1}{12}$$

$$\frac{5}{4}+\frac{7}{6}-\frac{9}{8}-\frac{13}{12}=*$$

$$*=\frac{30+28-27-26}{24}$$

$$*=\frac{58-53}{24}$$

$$*=\frac{5}{24}$$

120. Simplify : $\dfrac{121}{3\frac{2}{3}}+\dfrac{92}{7\frac{1}{3}}$

(a) $41\frac{12}{13}$ (b) $40\frac{13}{11}$

(c) $43\frac{11}{19}$ (d) $45\frac{6}{11}$

RRB JE - 23/05/2019 (Shift-II)

Ans : (d) From given expression,

$$\frac{121}{3\frac{2}{3}}+\frac{92}{7\frac{1}{3}}=\frac{121\times3}{11}+\frac{92\times3}{22}$$

$$=\frac{6\times121+92\times3}{22}=\frac{726+276}{22}$$

$$=\frac{1002}{22}=\frac{501}{11}=45\frac{6}{11}$$

121. Solve: $\dfrac{3.6\times0.48\times2.50}{0.12\times0.09\times0.5}$

(a) 8000 (b) 8

(c) 800 (d) 80

RRB JE - 24/05/2019 (Shift-I)

Ans : (c) From given expression,

$$\frac{3.6\times0.48\times2.50}{0.12\times0.09\times0.5}=\frac{36\times48\times250}{12\times9\times5}=16\times50=800$$

122. Find the value of

$$\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\frac{1}{4\times5}+\frac{1}{5\times6}+.....+\frac{1}{9\times10}$$

(a) 1/10 (b) 5/11

(c) 9/10 (d) 2/5

RRB JE - 24/05/2019 (Shift-III)

Ans : (c) From given expression,

$$\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\frac{1}{4\times5}+\frac{1}{5\times6}+.....+\frac{1}{9\times10}$$

$$\left[\frac{1}{1}-\frac{1}{2}\right]+\left[\frac{1}{2}-\frac{1}{3}\right]+\left[\frac{1}{3}-\frac{1}{4}\right]+\left[\frac{1}{4}-\frac{1}{5}\right]+\left[\frac{1}{5}-\frac{1}{6}\right]+.....+\left[\frac{1}{9}-\frac{1}{10}\right]$$

$$\left[\frac{10-1}{10}\right]=\frac{9}{10}$$

123. Simplify :

$$3+\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{4}}}$$

(a) 48/13 (b) 18/49

(c) 1/12 (d) 3/13

RRB JE - 25/05/2019 (Shift-I)

Ans : (a) From given expression,

$$=3+\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{4}}} \qquad =3+\cfrac{1}{1+\cfrac{1}{\frac{9}{4}}}$$

$$=3+\cfrac{1}{1+\cfrac{4}{9}} \qquad =3+\frac{9}{13}$$

$$=\frac{48}{13}$$

124. Simplify : 1.2×2.5 × 0.5

(a) 1.5 (b) 150

(c) 0.15 (d) 15

RRB JE - 25/05/2019 (Shift-II)

Ans : (a) From given expression,

$1.2\times2.5\times0.5$

$$=\frac{12\times25\times5}{10\times10\times10}=\frac{1500}{1000}=\boxed{1.5}$$

125. Simplify : $1.\overline{2}-0.1\overline{2}$

(a) 11/10 (b) 11/90

(c) 1 (d) 9/10

RRB JE - 26/05/2019 (Shift-III)

Ans : (a) From given expression,

$1.\overline{2}-0.1\overline{2}$

$=1+0.\overline{2}-0.1\overline{2}$

$$=1+\frac{2}{9}-\frac{(12-1)}{90}, \qquad =1+\frac{2}{9}-\frac{11}{90}$$

$$=1+\frac{20-11}{90}=1+\frac{9}{90}, \qquad =1+\frac{1}{10}=\frac{11}{10}$$

126. Simplify: $\left[1+\dfrac{1}{10+\frac{1}{10}}\right]+\left[1-\dfrac{1}{10+\frac{1}{10}}\right]$

(a) 2 (b) 101/10

(c) 3/10 (d) 91/101

RRB JE - 28/05/2019 (Shift-I)

Ans : (a) From given expression,

$$\left[1+\frac{1}{10+\frac{1}{10}}\right]+\left[1-\frac{1}{10+\frac{1}{10}}\right]$$

$$\left[1+\frac{10}{101}\right]+\left[1-\frac{10}{101}\right]$$

$$\left[\frac{101+10}{101}\right]+\left[\frac{101-10}{101}\right]$$

$$\frac{111}{101}+\frac{91}{101}=\frac{202}{101}=2$$

127. Simplify:

$$\frac{1\frac{1}{4}\div 1\frac{1}{2}}{\frac{1}{15}+1-\frac{9}{10}}$$

(a) 4 (b) 5
(c) 2 (d) 3

RRB JE - 30/05/2019 (Shift-II)

Ans : (b) From given expression,

$$\frac{1\frac{1}{4}\div 1\frac{1}{2}}{\frac{1}{15}+1-\frac{9}{10}}$$

$$\Rightarrow \frac{\frac{5}{4}\div\frac{3}{2}}{\frac{2+30-27}{30}}$$

$$\Rightarrow \frac{\frac{5}{4}\times\frac{2}{3}}{\frac{5}{30}}=\frac{\frac{5}{6}}{\frac{5}{30}}$$

$$=\frac{5}{6}\times\frac{30}{5}=5$$

128. Simplify :

$$0.\overline{09}\times 7.\overline{3}$$

(a) 11/3 (b) 61/9
(c) 2/3 (d) 67/99

RRB RPF-SI -06/01/2019 (Shift-III)

Ans : (c) From given expression,

$$0.\overline{09}\times 7.\overline{3}$$

$$=\frac{9}{99}\times\left(7+\frac{3}{9}\right)=\frac{9}{99}\times\frac{66}{9}=\frac{66}{99}=\frac{2}{3}$$

129. Solve: $\dfrac{1}{1+\dfrac{1}{1+\dfrac{1}{1-\frac{1}{2}}}}$

(a) 3/8 (b) 3/4
(c) 3/2 (d) 4/3

RRB JE - 31/05/2019 (Shift-I)

Ans : (b) From given expression,

$$\frac{1}{1+\dfrac{1}{1+\dfrac{1}{1-\frac{1}{2}}}}=\frac{1}{1+\dfrac{1}{1+\dfrac{1}{\frac{1}{2}}}}=\frac{1}{1+\dfrac{1}{1+\frac{2}{1}}}$$

$$=\frac{1}{1+\frac{1}{3}}=\frac{1}{\frac{3+1}{3}}=\frac{1}{\frac{4}{3}}=\frac{3}{4}$$

130. Simplify :

$$3.\overline{36}-2.\overline{05}+1.\overline{33}$$

(a) $2.\overline{61}$ (b) $2.\overline{64}$
(c) $2.\overline{64}$ (d) 2.64

RRB JE - 02/06/2019 (Shift-I)

Ans : (c) From given expression,

$$3.\overline{36}-2.\overline{05}+1.\overline{33}$$

$$=4.\overline{69}-2.\overline{05}=2.\overline{64}$$

131. Solve the following equation:

$$(7.5\times 7.5-2.5\times 2.5)\div(1.5^2+2.75)=?$$

(a) 10 (b) 50
(c) 20 (d) 5

RRB RPF-SI -05/01/2019 (Shift-III)

Ans. (a) : Given equation,

$$(7.5\times 7.5-2.5\times 2.5)\div(1.5^2+2.75)=?$$

$$=\frac{(7.5-2.5)(7.5+2.5)}{(2.25+2.75)}$$

$$=\frac{5.0\times 10.0}{5.00}$$

$$=\frac{5\times 10}{5}=10$$

132. Solve the following equation:

$$11\times 3\div\frac{1}{9}-5\times 6\left(1\div\frac{1}{6}\right)=?$$

(a) -115 (b) 117
(c) -117 (d) 115

RRB RPF-SI -13/01/2019 (Shift-II)

Ans : (b) From given expression,

$$11\times 3\div\frac{1}{9}-5\times 6\left(1\div\frac{1}{6}\right)$$

$$=11\times 3\times 9-5\times 6(1\times 6)$$

$$=297-180=117$$

133. 131/255 + 2/5 + 5 - 17/3 = ?

(a) $\frac{121}{255}$ (b) $\frac{117}{255}$
(c) $\frac{63}{255}$ (d) $\frac{131}{255}$

RRB RPF Constable -19/01/2019 (Shift-III)

Ans : (c) From given expression,

$$\frac{131}{255}+\frac{2}{5}+5-\frac{17}{3}$$

$$=\frac{131+102+1275-1445}{255}=\frac{63}{255}$$

134. $\frac{0.12\div0.15}{2}=?$

(a) 4 (b) 0.04
(c) 0.004 (d) 0.4

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (d) From given expression,

$$\frac{0.12\div0.15}{2}=?$$

$$\frac{0.12\times\frac{1}{0.15}}{2}=?$$

$$\frac{12}{15\times2}=?$$

$$?=0.4$$

135. Find the value of $\frac{7}{11}+\frac{8}{17}-\frac{1}{13}\times\frac{286}{11}$.

(a) $\frac{-167}{187}$ (b) $\frac{-161}{187}$
(c) $\frac{20}{187}$ (d) $\frac{-35}{187}$

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (a) From given expression,

$$\frac{7}{11}+\frac{8}{17}-\frac{1}{13}\times\frac{286}{11}$$

$$=\frac{7}{11}+\frac{8}{17}-\frac{286}{143}$$

$$=\frac{7}{11}+\frac{8}{17}-2, \qquad =\frac{119+88}{187}-2$$

$$=\frac{207}{187}-2, \qquad =\frac{207-374}{187}$$

$$=\frac{-167}{187}$$

136. $\frac{3-0.2}{0.1\times(3+0.2)}=?$

(a) 8.75 (b) 0.0875
(c) 87.5 (d) 0.875

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (a) From given expression,

$$\frac{3-0.2}{0.1\times(3+0.2)}$$

$$\Rightarrow\frac{2.8}{0.32}=8.75$$

137. Solve the following:

$$\frac{(0.54\times0.540-0.460\times0.460)}{(1-0.920)}=?$$

(a) 0.1 (b) 2
(c) 1 (d) 0.01

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c) From given expression,

$$\frac{0.54\times0.540-0.460\times0.460}{(1-0.920)}$$

$$(a^2-b^2)=(a+b)(a-b)$$

$$\Rightarrow\frac{(0.54)^2-(0.460)^2}{(1-0.920)}=\frac{(0.54+0.460)(0.54-0.460)}{(1-0.920)}$$

$$\Rightarrow\frac{1\times0.08}{0.08}=1$$

138. Find the value of the following equation:

$$\frac{1+2}{\left\{1+\frac{2}{1+\frac{1}{3}}\right\}}=?$$

(a) $\frac{21}{4}$ (b) $\frac{21}{5}$
(c) $\frac{6}{5}$ (d) $\frac{9}{4}$

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (c) From given expression,

$$\frac{1+2}{\left\{1+\frac{2}{\left(1+\frac{1}{3}\right)}\right\}}$$

$$=\frac{3}{\left\{1+\frac{2\times3}{4}\right\}}=\frac{3}{\frac{10}{4}}$$

$$=\frac{12}{10}=\frac{6}{5}$$

139. Solve the following equation:

$$11\div3+\frac{1}{9}-5\times6\left(1\times\frac{1}{6}\right)=?$$

(a) $\frac{-1}{9}$ (b) $\frac{-11}{9}$
(c) $\frac{11}{9}$ (d) $\frac{11}{8}$

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (b) From given expression,

$$11\div3+\frac{1}{9}-5\times6\left(1\times\frac{1}{6}\right)$$

$$=11\div3+\frac{1}{9}-5\times1$$

$$=\frac{11}{3}+\frac{1}{9}-5=\frac{33+1-45}{9}=\frac{-11}{9}$$

140. $\left(\frac{55}{11}\right)+(18-6)\times 9=?$

(a) 100 (b) 115
(c) 113 (d) 110

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (c) From given expression,

$\left(\frac{55}{11}\right)+(18-6)\times 9=?$

$5+12\times 9=?$

$5+108=?$

$?=113$

141. $(2.5)^2+8.7-\left(\frac{1.12}{1.4}\right)\times 0.5=?$

(a) 14.55 (b) 10.95
(c) 14.91 (d) 14.32

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (a) : From given expression,

$(2.5)^2+8.7-\left(\frac{1.12}{1.4}\right)\times 0.5=?$

$6.25+8.7-0.8\times 0.5=?$

$6.25+8.7-0.40=?$

$14.95-.40=?$

$\boxed{?=14.55}$

142. $\frac{0.16\times 1.65}{0.075\times 0.02^2}=?$

(a) 8400 (b) 8000
(c) 7500 (d) 8800

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : From given expression,

$?=\frac{0.16\times 1.65}{0.075\times 0.02^2}$

$=\frac{16\times 165\times 10^{-4}}{75\times 10^{-3}\times 4\times 10^{-4}}$

$=\frac{16\times 165\times 10^{3}}{75\times 4}$

$=\frac{4\times 165\times 10^{3}}{75}$

$=\frac{4\times 33\times 10^{3}}{15}$

$=\frac{4\times 33\times 10\times 100}{15}$

$=\frac{4\times 33\times 2\times 100}{3}$

$=4\times 11\times 2\times 100$

$=8800$

143. $\frac{3}{4}+\left\{\frac{3}{4}+\frac{3}{4}\div\left(\frac{3}{4}+\frac{3}{4}\right)\right\}=?$

(a) 1 (b) 2
(c) $\frac{3}{4}$ (d) $2\frac{3}{4}$

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (b) From given expression,

$\frac{3}{4}+\left\{\frac{3}{4}+\frac{3}{4}\div\left(\frac{3}{4}+\frac{3}{4}\right)\right\}$

$=\frac{3}{4}+\left\{\frac{3}{4}+\frac{3}{4}\div\frac{3}{2}\right\}$

$=\frac{3}{4}+\left\{\frac{3}{4}+\frac{1}{2}\right\}=\frac{3}{4}+\frac{3}{4}+\frac{1}{2}$

$=\frac{3+3+2}{4}=\frac{8}{4}=2$

144. $\frac{\left(\frac{2}{5}+\frac{4}{15}\right)}{\left(\frac{3}{5}-\frac{2}{5}\right)}$ of $\frac{3}{12}=?$

(a) $\frac{5}{7}$ (b) $\frac{2}{6}$
(c) $\frac{5}{6}$ (d) $\frac{6}{5}$

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (c) From given expression,

$\frac{\frac{2}{5}+\frac{4}{15}}{\left(\frac{3}{5}-\frac{2}{5}\right)}$ of $\frac{3}{12}=\frac{\frac{30+20}{75}}{\frac{1}{5}}\times\frac{3}{12}$

$\frac{50}{75}\div\frac{1}{5}\times\frac{3}{12}$

$\frac{50}{75}\times\frac{5}{1}\times\frac{3}{12}$

$\frac{50}{15}\times\frac{3}{12}=\frac{5}{6}$

145. Simplify: $\left(\frac{2}{3}\times\frac{4}{6}\right)+\left(\frac{5}{3}\times\frac{7}{2}\right)-\left(\frac{11}{4}\times\frac{4}{3}\right)$

(a) $\frac{31}{9}$ (b) $\frac{16}{9}$
(c) $\frac{47}{18}$ (d) $\frac{29}{18}$

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (c) From given expression,

$\left(\frac{2}{3}\times\frac{4}{6}\right)+\left(\frac{5}{3}\times\frac{7}{2}\right)-\left(\frac{11}{4}\times\frac{4}{3}\right)$

$=\frac{4}{9}+\frac{35}{6}-\frac{11}{3}$

$=\frac{8+105-66}{18}=\boxed{\frac{47}{18}}$

146. 2.09 ÷ 0.000209 = ?

(a) 100000 (b) 1000000
(c) 1000 (d) 10000

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (d) From given expression,

2.09 ÷ 0.000209

$= \frac{209}{100} \div \frac{209}{1000000}$

$= \frac{209}{100} \times \frac{1000000}{209}$

=100,00

147. 1.08 ÷ 0.000108 =

(a) 100000 (b) 1000
(c) 1000000 (d) 10000

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (d) : From given expression,

1.08 ÷ 0.000108

$= \frac{1.08}{0.000108}$

$= \frac{108}{0.0108}$

$= \frac{108 \times 10000}{108} = 10000$

148. Find the value of $3\frac{5}{8} + \frac{6}{16} - \frac{5}{24} + 3\frac{1}{2} = ?$

(a) $\frac{751}{24}$ (b) $\frac{715}{24}$
(c) $\frac{175}{24}$ (d) $\frac{157}{24}$

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (c) From given expression,

$3\frac{5}{8} + \frac{6}{16} - \frac{5}{24} + 3\frac{1}{2}$

$= \frac{29}{8} + \frac{6}{16} - \frac{5}{24} + \frac{7}{2}$

$= \frac{6 \times 29 + 3 \times 6 - 2 \times 5 + 24 \times 7}{48}$

$= \frac{174 + 18 - 10 + 168}{48} = \frac{350}{48} = \frac{175}{24}$

149. Simplify the equation:

$\left(\frac{5}{6} \times \frac{1}{3}\right) + \left(\frac{7}{3} \times \frac{1}{2}\right) - \left(\frac{11}{4} \times \frac{1}{3}\right) = ?$

(a) $\frac{19}{36}$ (b) $\frac{17}{9}$
(c) $\frac{14}{33}$ (d) $\frac{23}{9}$

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (a) : From given equation,

$= \left(\frac{5}{6} \times \frac{1}{3}\right) + \left(\frac{7}{3} \times \frac{1}{2}\right) - \left(\frac{11}{4} \times \frac{1}{3}\right)$

$= \frac{5}{18} + \frac{7}{6} - \frac{11}{12}$

$= \frac{20 + 84 - 66}{72}$

$= \frac{104 - 66}{72} = \frac{38}{72}$

$= \frac{19}{36}$

150. $\sqrt{75.24 + ?} = 8.71$

(a) 0.6241 (b) 6.0241
(c) 6.241 (d) 62.41

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (a) : From given expression,

$\sqrt{75.24 + ?} = 8.71$ Let ? = x

$\sqrt{75.24 + x} = 8.71$

On squaring the both side,

$(75.24 + x) = (8.71)^2$

75.24 + x = 75.8641

x = 75.8641 – 75.24

$\boxed{x = 0.6241}$

151. Solve the equation given below:

$\frac{(0.125 + 0.255)}{0.5 - 0.03 + 0.005} = ?$

(a) 0.08 (b) 0.8
(c) 0.2 (d) 1

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) : From given expression,

$\frac{(0.125 + 0.255)}{0.5 - 0.03 + 0.005}$

$= \frac{.380}{.475} = \frac{76}{95} = .8$

152. Solve:

(9.5 × 9.5 - 2.5 × 2.5) ÷ (1.5^2 - 0.25) = ?

(a) 47 (b) 42
(c) 49 (d) 45

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (b) From given expression,

(9.5 × 9.5 - 2.5 × 2.5) ÷ (1.5^2 - 0.25) = ?

$? = \frac{(9.5)^2 - (2.5)^2}{(1.5)^2 - (0.5)^2} = \frac{(9.5 - 2.5)(9.5 + 2.5)}{(1.5 + 0.5)(1.5 - 0.5)}$

$= \frac{7.0 \times 12.0}{2.0 \times 1.0} = 42.0$

153. 0.00025 ÷ 12.5 = ?

(a) 0.0025 (b) 0.00002
(c) 0.0002 (d) 0.000002

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (b) : 0.00025 ÷ 12.5

$= 0.00025 \times \frac{1}{12.5}$

$= \frac{0.00025}{12.5}, = \frac{25}{1250000} = 0.00002$

154. Simplify:

$$\left(\frac{7}{6}\times\frac{1}{3}\right)+\left(\frac{7}{3}\times\frac{3}{2}\right)-\left(\frac{13}{4}\times\frac{2}{3}\right)=?$$

(a) $\frac{31}{18}$ (b) $\frac{35}{18}$

(c) $\frac{36}{18}$ (d) $\frac{29}{18}$

RRB Group-D – 31/10/2018 (Shift-III)

Ans : (a) From given expression,

$$\left(\frac{7}{6}\times\frac{1}{3}\right)+\left(\frac{7}{3}\times\frac{3}{2}\right)-\left(\frac{13}{4}\times\frac{2}{3}\right)$$

$$=\frac{7}{18}+\frac{7}{2}-\frac{13}{6}$$

$$=\frac{7+63-39}{18}\Rightarrow\frac{70-39}{18}$$

$$=\frac{31}{18}$$

155. Solve the following equation:

$$\frac{16}{3}-\left\{4\frac{1}{3}-\left(3\frac{1}{3}-\left(2\frac{1}{3}-\frac{1}{3}\right)\right)\right\}=?$$

(a) $4\frac{1}{3}$ (b) $5\frac{1}{3}$

(c) $1\frac{1}{3}$ (d) $2\frac{1}{3}$

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) : From given expression,

$$\frac{16}{3}-\left\{4\frac{1}{3}-\left(3\frac{1}{3}-\left(2\frac{1}{3}-\frac{1}{3}\right)\right)\right\}=?$$

$$=\frac{16}{3}-\left\{4\frac{1}{3}-\left(3\frac{1}{3}-\left(\frac{7}{3}-\frac{1}{3}\right)\right)\right\}$$

$$=\frac{16}{3}-\left\{4\frac{1}{3}-\left(\frac{10}{3}-\frac{6}{3}\right)\right\}$$

$$=\frac{16}{3}-\left\{\frac{13}{3}-\frac{4}{3}\right\}$$

$$=\frac{16}{3}-\frac{9}{3},\quad =\frac{7}{3}\quad =2\frac{1}{3}$$

156. Find the value of $3.24\div0.002=?$

(a) 16.20 (b) 1620

(c) 162 (d) 1.620

RRB NTPC 26.04.2016 Shift : 3

Ans : (b) From given expression,

$3.24\div0.002$

$$\frac{3.24}{0.002}=\frac{324\times10}{2}=1620$$

157. (1 + 2/3) ÷ [(1+1/3) ÷ (2/3 +1)] = ?

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$

(c) $\frac{12}{25}$ (d) $\frac{25}{12}$

RRB NTPC 17.01.2017 Shift-2

Ans : (d) From given expression,

$$\left(1+\frac{2}{3}\right)\div\left[\left(1+\frac{1}{3}\right)\div\left(\frac{2}{3}+1\right)\right]$$

$$=\frac{5}{3}\div\left[\frac{4}{3}\times\frac{3}{5}\right]$$

$$=\frac{5}{3}\times\frac{5}{4}$$

$$=\frac{25}{12}$$

158. Find the value : $(0.3)^2\div100=?$

(a) 0.09 (b) 0.0009

(c) 0.009 (d) 0.9

RRB NTPC 17.01.2017 Shift-3

Ans : (b) From given expression,

$$(0.3)^2\div100=\frac{0.09}{100}$$

$= 0.0009$

159. Solve : $(0.25\times0.004)+0.374-0.72=?$

(a) -0.345 (b) 0.325

(c) 1.94 (d) -0.945

RRB NTPC 04.04.2016 Shift : 2

Ans : (a) $(0.25\times0.004)+0.374-0.72$

$=0.001-0.346$

$=-0.345$

160. Calculate: 0.00056÷0.07

(a) 0.08 (b) 0.008

(c) 0.0008 (d) 0.8

RRB NTPC 04.04.2016 Shift : 3

Ans : (b) From given expression,

$$0.00056\div0.07=\frac{0.00056}{0.07}\times\frac{100000}{100000}$$

$$=\frac{56}{7\times1000}=\frac{8}{1000}=0.008$$

161. Simplify: (3/2+5/3) ÷(3/2+2/3)

(a) 1 (b) 19/13

(c) 13/19 (d) 13/16

RRB NTPC 04.04.2016 Shift : 3

Ans : (b) From given expression,

$$\left(\frac{3}{2}+\frac{5}{3}\right)\div\left(\frac{3}{2}+\frac{2}{3}\right)=\left(\frac{9+10}{6}\right)\div\left(\frac{9+4}{6}\right)$$

$$=\frac{19}{6}\div\frac{13}{6}$$

$$=\frac{19}{6}\times\frac{6}{13}=\frac{19}{13}$$

162. Simplify : (2/5 + 2/9) ÷ (2/5 + 5/9)

(a) 28/45 (b) 28/43
(c) 27/34 (d) 7/17

RRB NTPC 02.04.2016 Shift : 1

Ans : (b) From given expression,

$$\left(\frac{2}{5}+\frac{2}{9}\right)\div\left(\frac{2}{5}+\frac{5}{9}\right)$$
$$=\left(\frac{18+10}{45}\right)\div\left(\frac{18+25}{45}\right)$$
$$=\frac{28}{45}\div\frac{43}{45}=\frac{28}{43}$$

163. Simplify: (2/9+3/5) ÷ (2/9+2/5).

(a) 37/28 (b) 47/43
(c) 43/47 (d) 41/47

RRB NTPC 02.04.2016 Shift : 3

Ans : (a) From given expression,

$$(2/9+3/5)\div(2/9+2/5)=\left(\frac{10+27}{45}\right)\div\left(\frac{10+18}{45}\right)$$
$$=\frac{37}{45}\div\frac{28}{45}$$
$$=\frac{37}{45}\times\frac{45}{28}=\frac{37}{28}$$

164. $5(10)^4+6(10)^3+4(10)-3(1/100)=?$

(a) 54,60.33 (b) 54,309.97
(c) 56,407.00 (d) 56,039.97

RRB NTPC 31.03.2016 Shift : 3

Ans : (d) From given expression,

$$5(10)^4+6\times(10)^3+4(10)-3(1/100)$$
$$=5\times10000+6\times1000+4\times10-3\times\frac{1}{100}$$
$$=50000+6000+40-.03$$
$$=56040-.03=56039.97$$

165. Find the value of 0.08 × 0.23 ÷ 0.004

(a) 46 (b) 4.6
(c) 0.046 (d) 0.0046

RRB NTPC 31.03.2016 Shift : 3

Ans : (b) From given expression,

$$0.08\times0.23\div0.004$$
$$=\frac{0.08\times0.23}{0.004}=\frac{8\times23}{4\times10}=\frac{2\times23}{10}=\frac{46}{10}=4.6$$

166. $\frac{0.55\times0.81}{4.5}=?$

(a) 0.99 (b) 0.099
(c) 9.9 (d) 0.0099

RRB NTPC 11.04.2016 Shift : 3

Ans : (b) From given expression,

$$\frac{0.55\times0.81}{4.5}$$
$$=\frac{55\times81\times10}{100\times100\times45}=0.099$$

167. Simplify:
(2/3 + 3/5) ÷ (2/3 + 2/5)

(a) 1 (b) 19/16
(c) 15/16 (d) 13/16

RRB NTPC 10.04.2016 Shift : 3

Ans : (b) From given expression,

$$\left(\frac{2}{3}+\frac{3}{5}\right)\div\left(\frac{2}{3}+\frac{2}{5}\right)$$
$$=\frac{10+9}{15}\div\frac{10+6}{15}=\frac{19}{15}\times\frac{15}{16}=\frac{19}{16}$$

168. Calculate : $\frac{40.7\times40.7\times40.7+1}{40.7\times40.7-40.7+1}$

(a) 417 (b) 4.17
(c) 41.7 (d) 441.7

RRB NTPC 09.04.2016 Shift : 3

Ans : (c) From given expression,

$$\frac{40.7\times40.7\times40.7+1}{40.7\times40.7-40.7+1}$$
$$=\frac{(40.7)^3+(1)^3}{(40.7)^2-40.7\times1+(1)^2}=(40.7+1)=40.7+1=41.7$$

Type - 3

169. Simplify : $\left(3+\sqrt{8}\right)+\frac{1}{(3-\sqrt{8})}-(6+4\sqrt{2})$

(a) 4 (b) 1
(c) 0 (d) 6

RRB JE - 23/05/2019 (Shift-III)

Ans : (c) From given expression,

$$=3+\sqrt{8}+\frac{1}{3-\sqrt{8}}\times\frac{3+\sqrt{8}}{3+\sqrt{8}}-\left(6+4\sqrt{2}\right)$$
$$=3+\sqrt{8}+3+\sqrt{8}-\left(6+4\sqrt{2}\right)$$
$$=6+2\sqrt{8}-6-4\sqrt{2}$$
$$=6+4\sqrt{2}-6-4\sqrt{2}$$
$$=0$$

170. Simplify : $\sqrt{25+10\sqrt{6}}+\sqrt{25-10\sqrt{6}}$

(a) $2\sqrt{15}$ (b) $2\sqrt{5}$
(c) $\sqrt{55}$ (d) $\sqrt{50}$

RRB JE - 24/05/2019 (Shift-III)

Ans : (a) From given expression,

$$\sqrt{25+10\sqrt{6}}+\sqrt{25-10\sqrt{6}}$$
$$X=\sqrt{25+10\sqrt{6}}+\sqrt{25-10\sqrt{6}}$$

On squaring both sides,

$$X^2=25+10\sqrt{6}+25-10\sqrt{6}+2\left(\sqrt{(25)^2-\left(10\sqrt{6}\right)^2}\right)$$
$$X^2=50+2\left(\sqrt{625-600}\right)$$

$X^2 = 50 + 2 \times 5$
$X^2 = 60$
$X = \sqrt{60}$
$X = 2\sqrt{15}$

171. If $3\sqrt{5} + \sqrt{125} = 17.88$, then find the value of $\sqrt{80} + 6\sqrt{5}$

(a) 22.25 (b) 22.35
(c) 18.75 (d) 20.235

RRB JE - 31/05/2019 (Shift-I)

Ans : (b) Given,

$3\sqrt{5} + \sqrt{125} = 17.88$
$3\sqrt{5} + 5\sqrt{5} = 17.88$
$8\sqrt{5} = 17.88$
$\sqrt{80} + 6\sqrt{5} = ?$
$4\sqrt{5} + 6\sqrt{5} = ?$
$10\sqrt{5} = ?$
$= \frac{17.88 \times 10\sqrt{5}}{8\sqrt{5}} = 22.35$

172. Find the value of $\sqrt{30 - \sqrt{30 - \sqrt{30 - \sqrt{30 - \sqrt{...}}}}}$?

(a) 5 (b) $\sqrt{30}$
(c) 5.4 (d) 6

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (a) Let

$\sqrt{30 - \sqrt{30 - \sqrt{30 - \sqrt{30 - \sqrt{.....}}}}} = x$

Squaring on both sides,

$30 - x = x^2$
$\Rightarrow x^2 + x - 30 = 0$
$\Rightarrow (x+6)(x-5) = 0$
$\Rightarrow x = 5$

173. Solve: $\frac{\sqrt{5}}{\sqrt{3} - \sqrt{2}} - \frac{3\sqrt{3}}{\sqrt{5} - \sqrt{2}} - \frac{\sqrt{8}}{\sqrt{5} + \sqrt{3}} = ?$

(a) 0 (b) 1/2
(c) -1 (d) -1/2

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (a) : From given expression,

$\frac{\sqrt{5}}{\sqrt{3} - \sqrt{2}} - \frac{3\sqrt{3}}{\sqrt{5} - \sqrt{2}} - \frac{\sqrt{8}}{\sqrt{5} + \sqrt{3}} = ?$

$? = \frac{\sqrt{5}(\sqrt{3} + \sqrt{2})}{3 - 2} - \frac{3\sqrt{3}(\sqrt{5} + \sqrt{2})}{5 - 2} - \frac{2\sqrt{2}(\sqrt{5} - \sqrt{3})}{5 - 3}$

$= \sqrt{15} + \sqrt{10} - \sqrt{15} - \sqrt{6} - \sqrt{10} + \sqrt{6}$
$= 0$

174. Solve the following:

$2\sqrt{12} \times 5\sqrt{20} \times 3\sqrt{15} = ?$

(a) 900 (b) 1800
(c) 1200 (d) 1500

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (b) From given expression,

$2\sqrt{12} \times 5\sqrt{20} \times 3\sqrt{15}$
$= 2\sqrt{2 \times 2 \times 3} \times 5\sqrt{2 \times 2 \times 5} \times 3\sqrt{15}$
$= 4\sqrt{3} \times 10\sqrt{5} \times 3\sqrt{15}$
$= 3 \times 40 \times \sqrt{15} \times \sqrt{15}$
$= 120 \times 15$
$= 1800$

Type - 3

175. Simplify: $\sqrt{12} \times \sqrt{27}$

(a) $4\sqrt{3}$ (b) $3\sqrt{4}$
(c) 18 (d) 9

RRB RPF-SI -06/01/2019 (Shift-III)

Ans : (c) From given expression,

$= \sqrt{12} \times \sqrt{27} = 2\sqrt{3} \times 3\sqrt{3} = 6 \times 3 = 18$

176. Find the value of 73×73+42×42-2×73×42

(a) 961 (b) 676
(c) 981 (d) 861

RRB RPF-SI -16/01/2019 (Shift-III)

Ans : (a) From given expression,

$73 \times 73 + 42 \times 42 - 2 \times 73 \times 42$
$= (73)^2 + (42)^2 - 2 \times 73 \times 42$
$= (73 - 42)^2 = (31)^2 = 961$

177. Find the value of $[(525+252)^2 - (525-252)^2]/(525 \times 252)$

(a) 3 (b) 4
(c) 5 (d) 6

RRB RPF Constable -22/01/2019 (Shift-III)

Ans : (b) From given expression,

$\frac{(525+252)^2 - (525-252)^2}{525 \times 252}$

$= \frac{(525)^2 + (252)^2 + 2 \times 525 \times 252 - (525)^2 - (252)^2 + 2 \times 525 \times 252}{525 \times 252}$

$= \frac{4 \times 525 \times 252}{525 \times 252} = 4$

178. Simplify: $\frac{0.1 \times 0.1 + 0.2 \times 0.2}{0.3 \times 0.3 + 0.6 \times 0.6}$

(a) 2/9 (b) 1/9
(c) 1/3 (d) 2/3

RRB JE - 27/05/2019 (Shift-I)

Ans : (b) From given expression,

$\frac{0.1 \times 0.1 + 0.2 \times 0.2}{0.3 \times 0.3 + 0.6 \times 0.6}$

$= \frac{\frac{1}{100} + \frac{4}{100}}{\frac{9}{100} + \frac{36}{100}} = \frac{5}{100} \times \frac{100}{45}$

$= \boxed{\frac{1}{9}}$

179. Find the value of : $\frac{(82+28)^2-(82-28)^2}{82\times 28}$

(a) 220 (b) 4
(c) 8 (d) 110

RRB JE - 28/06/2019 (Shift-III)

Ans. (B) Formula:

$\frac{(a+b)^2-(a-b)^2}{ab}=\frac{4ab}{ab}$ {a= 82, b=28}

From given expression,

$\because \frac{(82+28)^2-(82-28)^2}{82\times 28}$

$=\frac{4\times 82\times 28}{82\times 28}=4$

180. $(0.2\times 0.2\times 0.2)\times(0.06\times 0.06\times 0.06)\div(0.12\times 0.12\times 0.12)$=?

(a) 0.008 (b) 0.001
(c) 0.002 (d) 0.006

RRB RPF-SI -10/01/2019 (Shift-III)

Ans : (b) From given expression,

$\frac{(0.2)^3\times(0.06)^3}{(0.12)^3}$

$\frac{(0.2\times 0.06)^3}{(0.12)^3}=\left(\frac{0.012}{0.12}\right)^3$

$=\left(\frac{1}{10}\right)^3$

$=(0.1)^3$

$=0.001$

181. Solve: $\left(\sqrt{3}-\frac{1}{\sqrt{3}}\right)^2$

(a) 3/4 (b) 4/3
(c) $2\sqrt{3}$ (d) $\frac{4}{\sqrt{3}}$

RRB JE - 02/06/2019 (Shift-III)

Ans : (b) From given expression,

$\left(\sqrt{3}-\frac{1}{\sqrt{3}}\right)=\left(\sqrt{3}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2-2\sqrt{3}\times\frac{1}{\sqrt{3}}$

$=3+\frac{1}{3}-2=\frac{9+1-6}{3}=\frac{10-6}{3}=\frac{4}{3}$

182. Find the value of the following equation:

$\frac{(469+144)^2-(469-144)^2}{2(469\times 144)}=?$

(a) –2 (b) –1
(c) 1 (d) 2

RRB RPF Constable -19/01/2019 (Shift-II)

Ans : (d) Formula:- $\frac{(a+b)^2-(a-b)^2}{2ab}=\frac{4ab}{2ab}$

From given expression,

$\frac{(469+144)^2-(469-144)^2}{2(469\times 144)}=?$

$=\frac{(a+b)^2-(a-b)^2}{2(ab)}=\frac{4ab}{2ab}=2$

a = 469, b = 144

183. Solve the following equation:

$\frac{3^2-8^2}{(3+8)^2}=?$

(a) $\frac{3}{11}$ (b) $\frac{-5}{11}$
(c) $\frac{-3}{11}$ (d) $\frac{5}{11}$

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (b) From given expression,

$\frac{3^2-8^2}{(3+8)^2}=?$

$=\frac{(3+8)(3-8)}{(3+8)^2}$ $\left\{\begin{matrix}a^2-b^2=(a+b)(a-b)\\(a+b)^2=a^2+b^2+2ab\end{matrix}\right\}$

$=\frac{3-8}{3+8}$

$=\frac{-5}{11}$

184. Solve the equation: 467 × 467 + 166 × 166 –2 × 467 × 166 = ?

(a) 90106 (b) 960600
(c) 90601 (d) 90060

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (c) From given expression,

467 × 467 + 166 × 166 – 2 × 467 × 166

$(A–B)^2 = A^2 + B^2 – 2AB$

A = 467, B = 166

467 × 467 + 166 × 166 – 2 × 467 × 166

$(467)^2 + (166)^2 – (2 × 467 × 166)$

$(467 – 166)^2 = (301)^2 = 90601$

185. Solve: $\sqrt{(0.15+0.12)}\sqrt{(0.15-0.12)}$

(a) 0.09 (b) 0.03
(c) 0.9 (d) 0.3

RRB Group-D – 01/11/2018 (Shift-II)

Ans : (a) From given expression,

$\sqrt{(0.15+0.12)}\sqrt{(0.15-0.12)}$

$=\sqrt{(0.15)^2-(0.12)^2}$

$=\sqrt{0.0225-0.0144}$

$=\sqrt{0.0081}=0.09$

186. Solve the following:

$(3.6 + 6.4)(3.6 – 6.4) – (3.6 – 6.4)^2$ = ?

(a) 29.6 (b) –35.84
(c) 32.6 (d) 32.68

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (b) : From given expression,

$(3.6 + 6.4)(3.6 - 6.4) - (3.6 - 6.4)^2 = ?$

$\Rightarrow (3.6-6.4)[(3.6+6.4)-(3.6-6.4)] = ?$

$\Rightarrow (3.6-6.4)[3.6+6.4-3.6+6.4] = ?$

$\Rightarrow (3.6-6.4)(6.4+6.4) = ?$

$\Rightarrow (-2.8)(12.8) = ?$

$? = -35.84$

187. $(425)^2 - (424)^2 = ?$

(a) 859 (b) 869
(c) 839 (d) 849

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (d) From given expression,

$(425)^2 - (424)^2 = ?$

From formula- $a^2 - b^2 = (a + b)(a - b)$

$(425)^2 - (424)^2 = (425 + 424)(425 - 424)$

$= 849 \times 1$

$= 849$

188. The value of

$\{(.98)^3 + (0.02)^3 + 3 \times 0.98 \times 0.02 - 1\}$

(a) 1.09 (b) 1.98
(c) 0 (d) 1.562

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (c) : From given expression,

$\{(.98)^3 + (0.02)^3 + 3 \times 0.98 \times 0.02 - 1\}$

$\{(.98)^3 + (0.02)^3 + 3 \times 0.98 \times 0.02\,(.98 + .02) - 1\}$

$[(0.98)+(0.02)]^3 - 1$

$(1.00)^3 - 1 = 1 - 1 = 0$

189. $97 \times 97 = ?$

(a) 9391 (b) 9409
(c) 9049 (d) 9309

RRB NTPC 31.03.2016 Shift : 2

Ans : (b) From given expression,

$97 \times 97 = (97)^2$

$(100-3)^2 = (100)^2 + (3)^2 - 2 \times 100 \times 3$

$= 10000 + 9 - 600$

$= 9409$

190. $\left[\dfrac{1.93 \times 19.3 - 2.07 \times 20.7}{19.3 - 20.7}\right]$ equals to:

(a) 0.40 (b) 4.00
(c) 40 (d) 0.04

RRB NTPC 18.04.2016 Shift : 2

Ans : (b) From given expression,

$$\frac{1.93 \times 19.3 - 2.07 \times 20.7}{19.3 - 20.7}$$

$$= \frac{10(1.93 \times 1.93) - 10(2.07 \times 2.07)}{10(1.93 - 2.07)}$$

$$= \frac{(1.93)^2 - (2.07)^2}{(1.93 - 2.07)}$$

$$= \frac{(1.93 + 2.07)(1.93 - 2.07)}{(1.93 - 2.07)}$$

$= 1.93 + 2.07 = 4$

191. Find the value of following:

(250 × 250 × 250 + 150 × 150 × 150)/(250×250 – 250 × 150 + 150 × 150) = ?

(a) 40000 (b) 400
(c) 40 (d) 500

RRB NTPC 22.04.2016 Shift : 3

Ans : (b) From given expression,

$$= \frac{250 \times 250 \times 250 + 150 \times 150 \times 150}{250 \times 250 - 250 \times 150 + 150 \times 150}$$

$$= \frac{(250)^3 + (150)^3}{(250)^2 - 250 \times 150 + (150)^2}$$

$$\because \left[a^3 + b^3 = (a+b)(a^2 + b^2 - ab)\right]$$

$$\therefore = \frac{(250+150)\left[(250)^2 + (150)^2 - 250 \times 150\right]}{(250)^2 + (150)^2 - 250 \times 150}$$

$= 250 + 150 = 400$

192. Simplify : $(2.25)^{\frac{1}{2}}$

(a) 1.5 (b) 15
(c) 1.6 (d) 2/3

RRB NTPC 17.01.2017 Shift-3

Ans : (a) From given expression,

$(2.25)^{1/2}$

$= \sqrt{2.25}$

$$= \sqrt{\frac{225}{100}}$$

$$= \sqrt{\frac{15 \times 15}{10 \times 10}}$$

$$= \frac{15}{10} = 1.5$$

193. Simplify : $(7)^2 \div \left(7^{\frac{1}{2}}\right)^4$

(a) 1 (b) 7
(c) 14 (d) 1/14

RRB NTPC 19.01.2017 Shift : 1

Ans : (a) From given expression,

$7^2 \div 7^{\frac{1}{2} \times 4} = 7^2 \div 7^2 = 1$